

CSA0303 DATA STRUCTURES C01 AT3

OUTPUT PREDICTION SCREENSHOTS

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1) Merge two arrays

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, the subject DS PROGRAMMING, and the user's name and registration number. A sidebar on the left lists four questions, with the first one, 'Merging', selected. The main area shows the question details, a text prompt to write a C program, a code editor with a C program for merging two arrays, an input field with 'stdin input', and an empty output box. A 'Submitted' button is at the bottom.

QUESTIONS

- Q1** Merging 10 marks
- Q2** Reverse a string 10 marks
- Q3** Searching 10 marks
- Q4** Occurrence of an element 10 marks

4/4 answered

Q1 Merging 10 marks

Write a C program for Merge Two Arrays

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[10], b[10], c[20], n1, n2, i;
4     scanf("%d", &n1);
5     for (i=0; i<n1; i++)
6         scanf("%d", &a[i]);
7     scanf("%d", &n2);
8     for (i=0; i<n2; i++)
9         scanf("%d", &b[i]);
10    for (i=0; i<n1; i++)
11        c[i]=a[i];
12    for (i=0; i<n2; i++)
13        c[n1+i]=b[i];
14    for (i=0; i<n1+n2; i++)
15        printf("%d", c[i]);
16    return 0;
17 }
```

Input

stdin input

Output

Submitted

2) Reverse a sting (palindrome)

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QUESTIONS

Q1
Merging
10 marks

Q2
Reverse a string
10 marks

Q3
Searching
10 marks

Q4
Occurrence of an elemnet
10 marks

4/4 answered

Q2 Reverse a string 10 marks

Write a C program for palindrom

Your Code

```
1 #include<stdio.h>
2 int main() {
3     char s[30];
4     int i=0,j=0,f=1;
5     scanf("%s",s);
6     while(s[i]!='\0')
7         j++;
8     j--;
9     while(i<j) {
10         if(s[i]!=s[j])
11             f=0;
12         break;
13     }
14     i++;
15     j--;
16     if(f)
17         print("palindrome");
18     else
19         print("not a palindrome");
20     return 0;
21 }
```

Input

hello

Output

Security Error: Disallowed code pattern detected.

3) Searching an element in a sorted array

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QUESTIONS

Q1
Merging
10 marks

Q2
Reverse a string
10 marks

Q3
Searching
10 marks

Q4
Occurrence of an elemnet
10 marks

4/4 answered

Q3 Searching 10 marks

Write a C program for Searching an Element in a Sorted Array

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[]={10,20,30,40,50},x=30,i;
4     for(i=0;i<5;i++)
5         if(a[i]==x) {
6             printf("found");
7             return 0;
8         }
9     printf("not found");
10
11 }
```

Input

10 20 30 40 50
30

Output

found

Submitted

4) Occurrence of an element

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QUESTIONS

Q1
Merging
10 marks

Q2
Reverse a string
10 marks

Q3
Searching
10 marks

Q4
Occurrence of an elemnet
10 marks

4/4 answered

Q4 Occurrence of an elemnet 10 marks

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[10],n,x,i,f=4,l=8;
4     scanf("%d",&n);
5     for(i=0;i<n;i++);
6     scanf("%d",&a[i]);
7     scanf("%d",&x);
8     for(i=0;i<n;i++)
9     {
10        if(a[i]==x)
11        {
12            if(f==1)f=i;
13            l=i;
14        }
15    }
16    printf("%d %d",f,l);
17    return 0;
18 }
```

Input

4 8

Output

4 8

Run

Save

Submit